

# orbit\_demo verification summary

Package section 07\_verification of orbit\_demo\_design\_review, written 2026-10-06. Links are relative to this file. Paths in backticks without a link (reports/..., build/..., pd/..., ORFS image paths) are repository, build or tool-image paths. Byte-identical copies of the repository directories pd/, mk/, fault/, formal/, synth/, rtl/ and of part of scripts/ and docs/ are in [04\\_digital\\_design/source/](#) (cmp 2026-10-06); copies of reports/{formal,fault,synth,ecc,mutation} and of most of reports/pdsep/ are in this folder (section 2). build/... and ORFS image paths are not in the package unless section 2 lists a copy. Every result row carries one status label: VERIFIED (a tool log or report shows it), REPRODUCED 2026-10-06 (re-run for this package), TARGET (design goal, not measured), ASSUMPTION, CLAIM (unverified) (stated in a document, no log found), MISSING, N/A. Discrepancies are collected in the discrepancy register ([09\\_review\\_notes](#)); this document only flags them.

Scope: the reviewed layout is the copy-separated sky130hd run (variant sep, 7.2 ns). The baseline sky130hd run (variant base, 7.0 ns) is used only for comparison and is labelled as such. The ECC block is standalone and not in the layout (section 15).

## 1 Design revision covered by this package

Package revision table, verbatim from the package revision record:

| Item                 | Revision / identifier                                                                                                                                                                                                                                                                                                      | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Design               | orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)                                                                                                                                                                                                                                                                     | top module orbit_demo, LANES=4                                                                                                                                                                                                                                                                                                                                                                                                                            |
| RTL                  | rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26d71a8 (originally committed 2026-09-29 18:54 UTC; see the note on commit dates below); unchanged through the package commit                                                                                                      | sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)                                                                                                                                                                                                                                                                                                                                         |
| Specification        | docs/SPEC.md at 26d71a8 (never revised)                                                                                                                                                                                                                                                                                    | sha256 aef5d9fb...                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Concept brief        | docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)                                                                                                                                                                           | sha256 9d5f33d3...                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Reviewed layout      | ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/                                                                                                                                                                                 | 6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...                                                                                                                                                                                                                                                                                                                                                  |
| Layout configuration | pd/sky130hd_sep/config.mk and regions.tcl at 5e54769; constraint.sdc at 3e8dad1 (set clk_period 7.2)                                                                                                                                                                                                                       | The run (21:56 UTC) used config.mk/regions.tcl as in d8ead98, identical to 5e54769 once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which is the version in commit 3e8dad1 (originally committed at 22:00 UTC, after the run had started) and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc (written 21:56 UTC). d8ead98's constraint.sdc has a 7.0 ns period, so a rebuild must not use it |
| Flow / tools         | ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics) | No PDK commit is recorded in the logs                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Process / library    | SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)                                                                                                                                                                                                               | —                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Package assembled    | 2026-10-06; design inputs copied from the repository tree at commit 3dffffb (branch claudes/hopeful-rubin-0yf8io)                                                                                                                                                                                                          | Every later commit on the branch changes only review/ or the repository landing page README.md (added after packaging; not a design input), or is empty (git diff --name-only 3dffffb..HEAD outside review/ lists only README.md). Re-runs made for this package are dated 2026-10-06                                                                                                                                                                     |

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Note on commit dates: the commit identifiers above are on the current branch and have the same file trees as the commits the package was built from ([COMMIT\\_MAP.tsv](#) lists them with tree hashes). All dates in this package come from the tool logs and run records, which are dated 2026-09-29 to 2026-10-06, except times marked "originally committed", which are the commit times of the commits the package was built from (same period). The author and committer dates in the branch's commit metadata do not match these and should not be read as the development timeline (D-54).

Checked for this section on 2026-10-06: MANIFEST.sha256 is at the package root ([./MANIFEST.sha256](#), generated last on 2026-10-06; register D-53). The fault summary [fault/summary.md](#) lines 15-20 records the full sha256 of the four RTL files, and they match the table prefixes. The SPEF copy [6\\_final.spef.gz](#) decompresses to sha256 014e655ae938245a..., the value in the table. git diff --name-only 3dffffb..HEAD outside review/ was empty (checked 2026-10-06 at HEAD 78156c3); since then it lists only the repository-root README.md, added after packaging (not a design input). No design, report or flow file changed after the package commit.

## 2 Evidence in this folder

Unless a source cell names another path, the original of a package file is the folder origin below plus the same file name.

| Package folder                              | Contents                                                                           | Origin (repository or build path)             | How it got here                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">orfs_logs/</a>                  | ORFS step logs and metrics JSON of the sep run                                     | build/pd_sep/logs/sky130hd/orbit_demo/sep/    | copied before this section; 39 of 40 files, byte-identical (cmp 2026-10-06); the 40th, 5_1_grt.json, is in <a href="#">additional_evidence/flow/</a>                                                 |
| <a href="#">orfs_reports/</a>               | 6_finish.rpt, 6_drc.lyrdb, 6_drc_count.rpt, synth_stat/check, stage timing reports | build/pd_sep/reports/sky130hd/orbit_demo/sep/ | copied before this section; with 06_physical_design/orfs_images/ 23 of 25 files, byte-identical; the other two (drt_antennas.log, grt_antennas.log) are in <a href="#">additional_evidence/flow/</a> |
| <a href="#">published_reports/</a>          | results.md, checks.json, storage_audit.txt, power, paths, separation               | reports/pdsep/                                | copied before this section; 18 of 19 files, byte-identical                                                                                                                                           |
| <a href="#">gls/</a>                        | post-route GLS log and injection list                                              | build/pd_sep/gls/sep/                         | copied before this section; byte-identical                                                                                                                                                           |
| <a href="#">formal_rerun_2026-10-06/</a>    | make formal (12 tasks) and make formal-vacuity re-run                              | review scratch build                          | re-run 2026-10-06                                                                                                                                                                                    |
| <a href="#">sta_corners_sep_2026-10-06/</a> | multi-corner STA of the sep layout, scripts and logs                               | review scratch                                | new run 2026-10-06                                                                                                                                                                                   |
| <a href="#">formal/</a>                     | summary.md, vacuity.md, logs/ (13 tasks, 19 vacuity mutants)                       | reports/formal/                               | copied by this section (cp, not moved)                                                                                                                                                               |

| Package folder                       | Contents                                                                                                                                                                                                                                                                                                                                                                | Origin (repository or build path)                                                                                                                                                                                                                                                          | How it got here                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <a href="#">fault/</a>               | summary.md, campaign.md, campaign_per_bit.tsv, injection/, logs/                                                                                                                                                                                                                                                                                                        | reports/fault/                                                                                                                                                                                                                                                                             | copied by this section                                                                                        |
| <a href="#">synth/</a>               | generic synthesis, audits, negative control, equivalence, GLS                                                                                                                                                                                                                                                                                                           | reports/synth/                                                                                                                                                                                                                                                                             | copied by this section                                                                                        |
| <a href="#">ecc/</a>                 | results.txt, synth_stat.txt, logs/                                                                                                                                                                                                                                                                                                                                      | reports/ecc/                                                                                                                                                                                                                                                                               | copied by this section                                                                                        |
| <a href="#">mutation/</a>            | summary.md, results.json, tb_mutation_gaps.v                                                                                                                                                                                                                                                                                                                            | reports/mutation/                                                                                                                                                                                                                                                                          | copied by this section                                                                                        |
| <a href="#">baseline_reference/</a>  | sta_corners_baseline_7p0ns.txt; negctl_cfgkeep_audit.txt, negctl_nokeep_audit.txt (baseline ORFS keep controls); results_baseline_7p0ns.md                                                                                                                                                                                                                              | reports/pd/skyl30hd/: sta_corners.txt, negctl_*_audit.txt, results.md                                                                                                                                                                                                                      | copied by this section; sta_corners.txt and results.md renamed                                                |
| <a href="#">lec_inputs/</a>          | 4_rsz Lec_test.yml, 6_final_ Lec_test.yml (kepler-formal test configurations; they name the compared netlists, which are in <a href="#">additional_evidence/lec_netlists/</a> )                                                                                                                                                                                         | build/pd_sep/objects/skyl30hd/orbit_demo/sep/                                                                                                                                                                                                                                              | copied by this section                                                                                        |
| <a href="#">lvs/</a>                 | orbit_demo_extracted.cir.gz (layout-extracted netlist from KLayout LVS)                                                                                                                                                                                                                                                                                                 | build/pd_sep/results/ skyl30hd/orbit_demo/sep/                                                                                                                                                                                                                                             | gzip copy of orbit_demo_extracted.cir by this section (decompresses byte-identical)                           |
| <a href="#">rerun_2026-10-06/</a>    | re-run logs: make fault, make formal-extra, a second make formal-vacuity, make synth, make ecc, ECC sec_bmc sanity, sep storage audit with the current script, final STA/power/IR/EM re-query, GLS re-run, GLS netlist-mutant control                                                                                                                                   | review scratch review_work/ (formal_fault, synth, ecc, pd_sep, pd_ref)                                                                                                                                                                                                                     | copied by this section; logs contain scratch paths                                                            |
| <a href="#">additional_evidence/</a> | lec_netlists/: 1_synth_ Lec.v, 4_before_rsz_ Lec.v, 4_after_rsz_ Lec.v, 6_final_ Lec.v (the netlists kepler-formal compared). flow/: make_pd_sep.out (console log of the reviewed run, 7931 lines), 5_1_grt.json, drt_antennas.log and grt_antennas.log (0 bytes), 6_lvs.lvsdb (KLayout LVS database, 17.7 MB uncompressed), 6_final_concat.cd1 (LVS reference netlist) | build/pd_sep/results/skyl30hd/orbit_demo/sep/ (*_ Lec.v, 6_lvs.lvsdb); build/pd_sep/make_pd_sep.out; build/pd_sep/logs/skyl30hd/orbit_demo/sep/5_1_grt.json; build/pd_sep/reports/skyl30hd/orbit_demo/sep/ (antenna logs); build/pd_sep/objects/skyl30hd/orbit_demo/sep/6_final_concat.cd1 | added at the final step, 2026-10-06; gzip copies (the two 0-byte logs as is), decompress byte-identical (cmp) |

Not copied (size): formal and fault counterexample/cover VCDs under build/formal/ and build/fault/, per-trial campaign log build/fault/campaign/trials.tsv. Also not copied: reports/pdsep/orbit\_demo\_sky130hd\_sep.gds.gz (decompresses to the same sha256 cd19afa9... as [6\\_final.gds.gz](#)). Added at the final step (2026-10-06; register D-99), see [additional\\_evidence/](#) above: build/pd\_sep/results/skyl30hd/orbit\_demo/sep/6\_lvs.lvsdb (17.7 MB), build/pd\_sep/make\_pd\_sep.out (console log, 7931 lines), 5\_1\_grt.json (global-route metrics, 586912 B) from the ORFS log folder, drt\_antennas.log and grt\_antennas.log (both 0 bytes) from the ORFS report folder, 6\_final\_concat.cd1, and the LEC netlists 1\_synth\_ Lec.v, 6\_final\_ Lec.v, 4\_before\_rsz\_ Lec.v, 4\_after\_rsz\_ Lec.v from build/pd\_sep/results/skyl30hd/orbit\_demo/sep/. With these and [lec\\_inputs/](#) the LEC can be re-run from the package; the yml files name the build paths, and the TT liberty and kepler-formal come from the ORFS image. 6\_final.odt is in [06\\_physical\\_design/layout\\_db/](#).

## 3 Tools, conditions, targets and assumptions

### 3.1 Tools

| Tool                                                                   | Version as recorded                                                                                                                                           | Used for                          | Source                                                                             | Status                                               |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|
| OpenROAD (OpenSTA, TritonRoute, TritonCTS, OpenRCX, PSM, ANT)          | prints "unknown"; build features -GPU +GUI -Python                                                                                                            | P&R, STA, extraction, IR, antenna | <a href="#">6_report.log</a> line 1; SPEF header *VERSION "unknown"                | VERIFIED                                             |
| ORFS docker image                                                      | tag openroad/orfs:latest; local image digest sha256:2e5bf6fe865e..., created 2026-09-29 02:00 UTC                                                             | whole flow                        | no log records the digest; only one ORFS image on the host                         | ASSUMPTION (that this image ran the 2026-09-29 flow) |
| ORFS commit, PDK (open_pdk / skywater-pdk) commit                      | not recorded                                                                                                                                                  | —                                 | —                                                                                  | MISSING                                              |
| Yosys (ORFS synthesis)                                                 | 0.68+post (git sha1 UNKNOWN)                                                                                                                                  | sky130hd mapping                  | <a href="#">1_2_yosys.log</a> line 1                                               | VERIFIED                                             |
| KLayout                                                                | 0.30.12                                                                                                                                                       | DRC, LVS, GDS merge               | <a href="#">6_drc.log</a> line 1; <a href="#">6_lvs.log</a> line 1                 | VERIFIED                                             |
| kepler-formal                                                          | version not printed; tool name only in ORFS flow/scripts/formal_check.tcl and <a href="#">pd/collect_pd.py</a> line 150                                       | LEC                               | <a href="#">6_final_ Lec_check.log</a>                                             | MISSING (version)                                    |
| Yosys / SBY / Yices / bitwuzla (formal, fault, generic synthesis, ECC) | Yosys 0.69+154 (git sha1 30d62572e-dirty); SBY v0.69; Yices 2.7.0; bitwuzla 0.9.1; abc pdr and aiger suprove from the same OSS CAD Suite, versions not stated | formal, fault, formal, synth, ECC | <a href="#">formal/summary.md</a> line 4; <a href="#">fault/summary.md</a> line 11 | VERIFIED                                             |
| Icarus Verilog                                                         | 14.0 (devel) (s20260301-500-g2e81fccb-dirty)                                                                                                                  | SEU campaign, GLS, ECC sim        | <a href="#">fault/summary.md</a> line 11                                           | VERIFIED                                             |
| Verilator                                                              | 5.053 devel rev v5.052-233-gf5f9ddef9                                                                                                                         | ECC lint                          | <a href="#">lint_verilator.log</a> line 1                                          | VERIFIED                                             |

### 3.2 Common timing / physical conditions

Unless stated: sep layout 6\_final.odt / [6\\_final.sdc](#) / [6\\_final.spef.gz](#); create\_clock clk -period 7.2000 (6\_final.sdc line 8), set\_propagated\_clock (line 9); set\_input\_delay 1.4400 on 79 inputs and set\_output\_delay 1.4400 on 135 outputs (0.2 x period); no set\_clock\_uncertainty, set\_timing\_derate, set\_load, set\_driving\_cell or set\_input\_transition in 6\_final.sdc (grep 2026-10-06). Liberty sky130\_fd\_sc\_hd\_tt\_025C\_1v80.lib (TT, 25 C, 1.80 V), the only liberty ORFS read ([6\\_report.log](#) line 6).

### 3.3 Targets and assumptions referenced in this section (not results)

| Item                                                 | Value                                                                                                                                                                                                                   | Source                                                  | Status                         |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------|
| Demonstrator clock requirement                       | none in SPEC.md; 7.2 ns is the shortest period tried that closed at TT ( <a href="#">period_exploration.md</a> )                                                                                                        | <a href="#">SPEC.md</a>                                 | MISSING (requirement)          |
| Concept chip: 16 tiles, 200-800 MHz, 6.55-26.21 TOPS | concept targets for a different, unbuilt chip (p.1 "unvalidated targets"; p.5 also calls the tile count and clock range project assumptions); they do not apply to orbit_demo and nothing in this section verifies them | <a href="#">brief</a> p.1-2, p.5                        | TARGET                         |
| Concept chip: 40 W per chip                          | conceptual sizing allocation, not a power estimate (p.1 "sizing assumption - not an estimate"; p.2 "for conceptual sizing only"); does not apply to orbit_demo                                                          | <a href="#">brief</a> p.1-2, p.5                        | ASSUMPTION                     |
| SPEC section 5 integrity target                      | an upset in one copy of accumulator/result storage never transfers a wrong out_data; it causes a stop                                                                                                                   | <a href="#">SPEC.md</a> lines 93-94                     | TARGET (checked in section 12) |
| I/O delay 20 % of period                             | placeholder for the unknown outside world                                                                                                                                                                               | <a href="#">constraint.sdc</a> lines 19-27              | ASSUMPTION                     |
| Fault model                                          | single upset per trace/trial; synchronous model, timing closure assumed, metastability out of scope                                                                                                                     | <a href="#">fault/summary.md</a> lines 33-41            | ASSUMPTION                     |
| Copy-separation acceptance limits                    | same-bit centre >= 20 um, edge gap >= 10 um; no rationale given                                                                                                                                                         | <a href="#">scripts/pdsep_separation.py</a> lines 46-48 | TARGET                         |

## 4 Physical verification

### 4.1 DRC

| Check                                                                 | Result                                                                                                      | Tool + version                                                                   | Conditions                                                                                                                                                                                                                                                                                                                                                                                                                           | Status                          | Source                                                                                                                                                                    |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detailed-route DRC                                                    | 0 violations after the last iteration (35 at the preceding report step); 5_route_drc.rpt is empty (0 bytes) | TritonRoute in OpenROAD "unknown"                                                | detailed_route -droute_end_iter 64; router rules from the sky130hd tech LEF. DRT-0349: LEF58_ENCLOSURE without CUTCLASS not supported, enclosure rules skipped for mcon, via, via2, via3, via4                                                                                                                                                                                                                                       | VERIFIED                        | <a href="#">5_2_route.log</a> lines 10-21, 973, 1007; <a href="#">5_route_drc.rpt</a>                                                                                     |
| KLayout DRC, ORFS deck                                                | 0 markers: 6_drc_count.rpt = 0; 6_drc.lyrdb has 145 rule categories and 0 items                             | KLayout 0.30.12, ORFS deck sky130hd.lydrc ("v1.0 : initial release", 2020-10-04) | Deck scope: FEOL = false, BEOL = true, OFFGRID = true (deck lines 46-48, read from the ORFS image). Log runs only "DRC section", "BEOL section", "OFFGRID-ANGLES section". Categories: li, ct, m1-m5, via-via4, nsm, pad.2, plus *_OFFGRID / *_angle. No FEOL rules (well, diff/tap, poly, licon, implants), no metal-density rules, no antenna or latch-up rules. Run on 6_final.gds without metal fill. This is NOT a sign-off DRC | VERIFIED                        | <a href="#">6_drc_count.rpt</a> ; <a href="#">6_drc.lyrdb</a> ; <a href="#">6_drc.log</a> lines 1, 257-258, 1222; command in build/pd_sep/make_pd_sep.out lines 5517-5519 |
| Metal fill                                                            | not inserted: step 6_1_fill only copies 5_route.odt to 6_1_fill.odt                                         | OpenROAD "unknown"                                                               | ORFS sky130hd platform default                                                                                                                                                                                                                                                                                                                                                                                                       | VERIFIED (that no fill was run) | <a href="#">6_1_fill.log</a> line 10                                                                                                                                      |
| Metal density on a filled GDS                                         | not run                                                                                                     | —                                                                                | needs fill and a density check. The tech LEF read by the flow defines MAXIMUMDENSITY 70 / DENSITYCHECKWINDOW 700 700 for met1-met4 (no minimum density); no step compared the layout with it                                                                                                                                                                                                                                         | MISSING                         | sky130_fd_sc_hd.tlef (ORFS image copy) lines 120-121, 165-166, 206-207, 247-248; read by the flow per <a href="#">1_synth.log</a> line 7 (ODB-0227)                       |
| FEOL DRC (full sky130 rule deck, e.g. open_pdk KLayout sign-off deck) | not run                                                                                                     | —                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                    | MISSING                         | —                                                                                                                                                                         |
| Magic DRC                                                             | not run; no Magic log or tech file use in the flow                                                          | —                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                    | MISSING                         | —                                                                                                                                                                         |

The PASS row in [results.md](#) line 20 ("KLayout DRC PASS, 0 markers") does not state the deck scope; see discrepancy register (09\_review\_notes).

### 4.2 LVS

| Check                                                                                    | Result                                    | Tool + version                            | Conditions                                                                                                                                                                                                                                                                                                                                              | Status   | Source                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layout vs schematic                                                                      | "INFO : Congratulations! Netlists match." | KLayout 0.30.12, ORFS deck sky130hd.lylvs | Layout: 6_final.gds extracted at device level (orbit_demo_extracted.cir). Reference: 6_final_concat.cdl = 6_final.cdl (written by OpenROAD from 6_final.odt) concatenated with the platform cell CDL sky130hd.cdl. The reference comes from the same database as the GDS, so this confirms stream-out and cell connectivity, not an independent netlist | VERIFIED | <a href="#">6_lvs.log</a> lines 1, 649, 652; <a href="#">6_cdl.log</a> ; reference <a href="#">6_final.cdl.gz</a> ; extracted <a href="#">orbit_demo_extracted.cir.gz</a> ; see note below |
| Independent LVS (Magic extraction + Netgen against 6_final.v or the synthesized netlist) | not run                                   | —                                         | —                                                                                                                                                                                                                                                                                                                                                       | MISSING  | —                                                                                                                                                                                          |

Note: the concatenation command is in build/pd\_sep/make\_pd\_sep.out line 7244 ([make\\_pd\\_sep.out.gz](#)); the reference netlist is [6\\_final\\_concat.cdl.gz](#); the LVS database is build/pd\_sep/results/sky130hd/orbit\_demo/sep/6\_lvs.lvldb (17.7 MB), packaged as [6\\_lvs.lvldb.gz](#). Deck paths are inside the ORFS image under /OpenROAD-flow-scripts/flow/.

### 4.3 Logic equivalence (LEC) inside the flow

| Check                                                  | Result                                                       | Tool + version                                          | Conditions                                                                                                                                                                                                           | Status   | Source                                                                                                                                                                                                                                      |
|--------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4_rsz LEC, published as "LEC synth vs post-CTS repair" | "Circuits are IDENTICAL"; 613 / 613 PIs, 1177 Pos, SAT UNSAT | kepler-formal (version MISSING), called by ORFS cts.tcl | Compares 4_before_rsz Lec.v with 4_after_rsz Lec.v: the netlists immediately before and after repair_timing inside the CTS step (ORFS flow/scripts/cts.tcl lines 61-71). It is NOT synthesis vs post-CTS. Liberty TT | VERIFIED | <a href="#">4_rsz Lec_check.log</a> lines 4, 6, 95; <a href="#">4_rsz Lec_test.yml</a> ; <a href="#">4_1_cts.log</a> line 469; compared netlists now included: <a href="#">4_before_rsz Lec.v.gz</a> , <a href="#">4_after_rsz Lec.v.gz</a> |

| Check                         | Result                                                              | Tool + version                  | Conditions                                                                                                                                                                                                                                                                                | Status   | Source                                                                                                                                                                                                      |
|-------------------------------|---------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6_final LEC, "synth vs final" | "Circuits are IDENTICAL"; 603 / 613 PIs, 1177 common POs, SAT UNSAT | kepler-formal (version MISSING) | Compares 1_synth_lec.v with 6_final_lec.v. Combinational miter with the 521 dfxtp_1 Q pins as cut points; 1177 POs = 521 CLK + 521 D + 135 output bits. The 10 extra PIs are tie-cell outputs (conb_1 HI/LO: 1+1 in synthesis, 6+6 in the final netlist). Does not cover RTL vs synthesis | VERIFIED | <a href="#">6_final_lec_check.log</a> lines 4, 6, 8-24, 44, 109; <a href="#">6_final_lec_test.yml</a> ; compared netlists now included: <a href="#">1_synth_lec.v.gz</a> , <a href="#">6_final_lec.v.gz</a> |

The label in [results.md](#) line 22 and [pd/collect\\_pd.py](#) line 151 misnames the 4\_rsz check; see discrepancy register (09\_review\_notes). RTL-to-gate equivalence is in section 13.

#### 4.4 Antenna

| Check                             | Result                                                                        | Tool + version                | Conditions                                            | Status   | Source                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------|
| Global-route antenna repair       | 31 antenna violations found, 0 after repair                                   | OpenROAD GRT / ANT, "unknown" | ORFS default antenna rules from the sky130hd tech LEF | VERIFIED | <a href="#">5_1_grt.log</a> lines 663-668                                                                                      |
| Detailed-route antenna repair     | 18 diodes inserted, then 2 more                                               | OpenROAD DRT / GRT-0015       | —                                                     | VERIFIED | <a href="#">5_2_route.log</a> lines 425, 765                                                                                   |
| Final antenna check               | 0 net violations, 0 pin violations; 20 antenna cells in the final cell report | OpenROAD ANT-0002 / ANT-0001  | routed database                                       | VERIFIED | <a href="#">5_2_route.log</a> lines 1056-1059; <a href="#">6_report.json</a> line 14; <a href="#">6_report.log</a> cell report |
| Antenna check by KLayout or Magic | not run: the KLayout deck has no antenna rule categories                      | —                             | —                                                     | MISSING  | <a href="#">6_drc.lyrdb</a> category list                                                                                      |

## 5 Extracted-parasitic results

| Check                                                    | Result                                                                                                                                                                     | Tool + version                | Conditions                                                                                                                                                                                                                                                                                 | Status   | Source                                                                                                                            |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Parasitic extraction                                     | 6886 nets, 30095 rsegs, 30095 caps, 60471 coupling caps; written to 6_final.spef (6827076 bytes, sha256 014e655a...) and read back for the final STA, power and IR reports | OpenRCX in OpenROAD "unknown" | Rules rcx_patterns.rules of the ORFS sky130hd platform (the only RCX rules file there); one process corner X, ext_model_index 0 (a single nominal extraction corner); coupling threshold 0.1 fF; max_merge_res 50.0. Provenance and qualification of rcx_patterns.rules are not documented | VERIFIED | <a href="#">6_report.log</a> lines 16-26; <a href="#">6_final.spef.gz</a> header; ORFS flow/scripts/final_outputs.tcl lines 27-37 |
| STA on the SPEF                                          | final ORFS STA, power and IR use the OpenRCX SPEF; the 2026-10-06 corner runs read the same SPEF (read_spef) at every liberty corner                                       | OpenSTA in OpenROAD "unknown" | see section 6                                                                                                                                                                                                                                                                              | VERIFIED | ORFS final_outputs.tcl line 37; <a href="#">sta_corners_extra.tcl</a> ; <a href="#">pd/sta_corners.tcl</a> lines 14-18            |
| RC corners (min/max extraction rules)                    | not available; nominal SPEF used at every PVT corner                                                                                                                       | —                             | —                                                                                                                                                                                                                                                                                          | MISSING  | <a href="#">pd/sta_corners.tcl</a> lines 9-10                                                                                     |
| Post-layout SPICE / transistor-level simulation          | not run (the LVS-extracted device netlist exists but carries no parasitics and was never simulated)                                                                        | —                             | —                                                                                                                                                                                                                                                                                          | MISSING  | —                                                                                                                                 |
| Post-layout timing (SDF-annotated) gate-level simulation | not run; every GLS in the package is zero-delay                                                                                                                            | —                             | —                                                                                                                                                                                                                                                                                          | MISSING  | <a href="#">tb_gls.v</a> header lines 12-13                                                                                       |

## 6 Timing

### 6.1 ORFS final STA at TT (the flow's own final report; single corner)

| Check                                 | Result                                                                                                                                                                                                  | Tool + version                                           | Conditions                                                              | Status                | Source                                                                                                                                                                            |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Setup                                 | WNS +0.0310081 ns, TNS 0, 0 setup violations                                                                                                                                                            | OpenSTA in OpenROAD "unknown"                            | section 3.2; TT 25 C 1.80 V; OpenRCX SPEF                               | VERIFIED              | <a href="#">6_report.json</a> lines 30, 32; <a href="#">6_finish.rpt</a> lines 5, 15, 416                                                                                         |
| Worst setup path                      | in_a[27] (input, 1.44 ns external delay) -> g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_.slack 0.03 ns (MET); an I/O path that crosses the die from the east-edge pin to the copy-A fence               | same                                                     | same                                                                    | VERIFIED              | <a href="#">worst_setup_path.txt</a> lines 1-2, 125; <a href="#">6_final.def.gz</a> : pin in_a[27] PLACED (345895 109140) dbu, flip-flop PLACED (43700 108800) dbu (0.001 um/dbu) |
| Hold                                  | WNS +0.435959 ns, TNS 0, 0 hold violations; worst path u_thermal.phase\$_SDFF_PN0_ -> itself, slack 0.44                                                                                                | same                                                     | same                                                                    | VERIFIED              | <a href="#">6_report.json</a> lines 31, 33; <a href="#">worst_hold_path.txt</a> line 53; <a href="#">6_finish.rpt</a> line 421                                                    |
| Clock skew                            | setup skew 0.0538434 ns (latency 0.65 at u_thermal.u_copy0/q[0]\$_SDFFE_PN0P_/CLK, -0.59 at g_lane[3].u_lane.u_res_b/q[20]\$_SDFFE_PN0P_/CLK); one TritonCTS H-tree for all 521 sinks, 56 clock buffers | OpenSTA / TritonCTS                                      | propagated clock                                                        | VERIFIED              | <a href="#">6_report.json</a> line 36; <a href="#">6_finish.rpt</a> lines 23-30; <a href="#">4_1_cts.log</a> lines 17-43                                                          |
| Min period / fmax incl. I/O paths     | 7.17 ns / 139.49 MHz (report_clock_min_period - include_port_paths, with the assumed 1.44 ns I/O delays held fixed)                                                                                     | OpenSTA                                                  | same                                                                    | VERIFIED              | <a href="#">6_finish.rpt</a> line 20; ORFS report_metrics.tcl line 39                                                                                                             |
| Min period reg-to-reg (no port paths) | 5.85 ns / 170.82 MHz; same-database re-query also returns setup WNS 0.031008 ns, hold WNS 0.435959 ns, 7.17 ns with port paths                                                                          | OpenSTA in OpenROAD "unknown", image sha256:2e5bf6fe865e | <a href="#">sta.tcl</a> : read 6_final.odb/sdc, setRC.tcl, 6_final.spef | REPRODUCED 2026-10-06 | <a href="#">sta.log</a> lines 5, 10-11                                                                                                                                            |
| Intermediate (estimate-based) timing  | did not close before the final RCX-based STA: RSZ-0062 Unable to repair all setup violations at floorplan, CTS and global route                                                                         | OpenROAD resizer                                         | placement/route estimates                                               | VERIFIED              | <a href="#">2_1_floorplan.log</a> line 362; <a href="#">4_1_cts.log</a> line 463; <a href="#">5_1_grt.log</a> line 620                                                            |

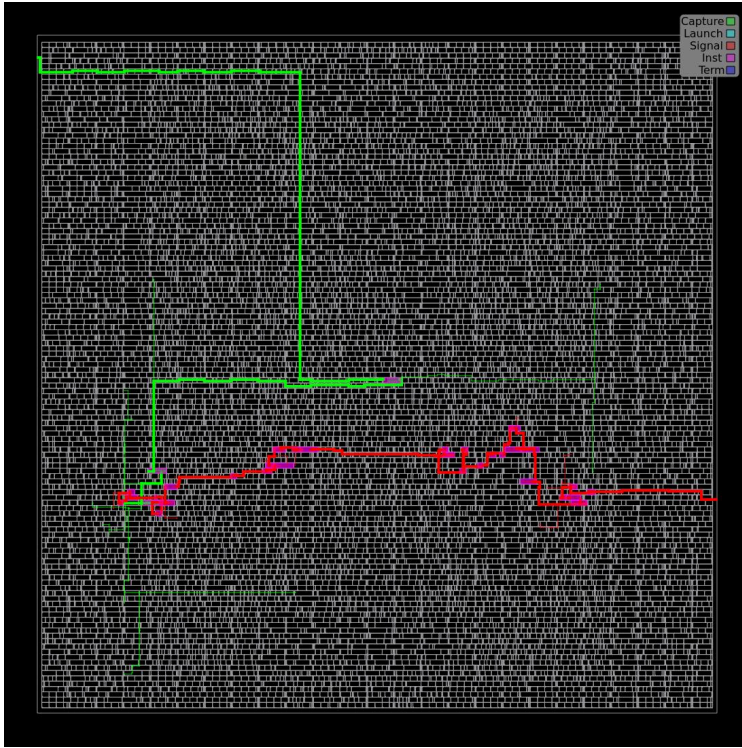


Figure 1: ORFS GUI export `final_worst_path` of the sep layout (TT, 7.2 ns): the worst setup path runs from the east-edge port `in_a[27]` to the copy-A fence at the west edge. Source [final\\_worst\\_path.webbp.png](#) (= `build/pd_sep/reports/sky130hd/orbit_demo/sep/final_worst_path.webbp.png`).

## 6.2 Multi-corner re-time of the sep layout (new, 2026-10-06)

Conditions for every row: `pd/sta_corners.tcl` unmodified, run through `scripts/run_pd.sh --shell` with `openroad -no_init -threads 1` in image `openroad/orfs:latest` (sha256:2e5bf6fe865e), inputs `build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.{odb, sdc, spef}`, 7.2000 ns, 1.44 ns I/O delay, propagated clock, nominal OpenRCX SPEF at every corner. The design was optimised by ORFS at TT only; SS and FF re-time the same routed database (information only). SS/FF liberty from the master branch of `efabless/skywater-pdk-libs-sky130_fd_sc_hd` (sha256 prefixes `ss 9b24f0db3967ac67`, `ff fb61d91c55a7f85b`; no commit pinned), reused from `build/pd/platform/sky130hd/corners/`. The script header line "Constraint: the committed SDC" is fixed text; the input is the ORFS-written `6_final.sdc` (same 7.2 ns); see discrepancy register (09\_review\_notes).

| Corner (liberty)               | Setup WNS / TNS (ns)                | Hold WNS / TNS (ns)                   | Min period / fmax (incl. I/O) | Worst setup path                                                         | Tool + version              | Status                              | Source                                                                  |
|--------------------------------|-------------------------------------|---------------------------------------|-------------------------------|--------------------------------------------------------------------------|-----------------------------|-------------------------------------|-------------------------------------------------------------------------|
| tt<br>(tt_025C_1v80)           | +0.031 / 0.000                      | +0.436 / 0.000                        | 7.17 ns / 139.49 MHz          | <code>in_a[27] -&gt; g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_</code> | OpenSTA, OpenROAD "unknown" | REPRODUCED 2026-10-06 (matches 6.1) | <a href="#">sta_corners_sep.txt</a> line 13; <a href="#">sta_tt.log</a> |
| ss_100C_1v60<br>(ss_100C_1v60) | -6.131 / -2463.532: FAILS at 7.2 ns | +0.891 / 0.000                        | 13.33 ns / 75.01 MHz          | <code>in_a[12] -&gt; g_lane[1].u_lane.u_acc_a/q[31]\$_SDFFE_PN0P_</code> | same                        | REPRODUCED 2026-10-06               | line 14; <a href="#">sta_ss_100C_1v60.log</a> lines 6-11                |
| ff_n40C_1v95<br>(ff_n40C_1v95) | +2.276 / 0.000                      | +0.281 / 0.000 (smallest hold margin) | 4.92 ns / 203.07 MHz          | <code>in_b[30] -&gt; g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_</code> | same                        | REPRODUCED 2026-10-06               | line 15; <a href="#">sta_ff_n40C_1v95.log</a>                           |

Supplementary queries ([sta\\_corners\\_extra.tcl](#), same setup as `pd/sta_corners.tcl`; [sta\\_corners\\_sep\\_supplement.txt](#), generated by `extract_supplement.py`):

| Check                               | Result                                                                                                                                                                                                                                                                                                                            | Tool + version | Conditions                                                                               | Status                | Source                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------|
| Worst setup slack by path class, TT | <code>reg2reg +1.346 (g_lane[0].u_lane.u_res_b/q[30]), in2reg +0.031, reg2out +0.654 (in_ready), in2out +2.577</code>                                                                                                                                                                                                             | OpenSTA        | <code>report_checks -from/-to class, -format end</code>                                  | REPRODUCED 2026-10-06 | supplement, corner tt                                                          |
| Worst setup slack by path class, SS | <code>reg2reg -4.420, in2reg -6.131, reg2out -4.142 (in_ready), in2out +1.195</code> . The SS failure is not only the I/O budget: <code>reg-to-reg</code> fails by 4.420 ns                                                                                                                                                       | OpenSTA        | same                                                                                     | REPRODUCED 2026-10-06 | supplement, corner ss_100C_1v60                                                |
| Worst setup slack by path class, FF | <code>reg2reg +3.571, in2reg +2.276, reg2out +2.558, in2out +3.165</code>                                                                                                                                                                                                                                                         | OpenSTA        | same                                                                                     | REPRODUCED 2026-10-06 | supplement, corner ff_n40C_1v95                                                |
| Setup-violating endpoints at SS     | 516: all 32 bits of each of the 16 registers <code>g_lane[0..3].u_lane.u_{acc_a, acc_b, res_a, res_b}</code> (512), <code>out_valid_q\$_SDFFE_PP0P_/D</code> , <code>fault_q\$_SDFFE_PP0P_/D</code> , outputs <code>in_ready</code> and <code>out_valid</code> . 0 at TT and FF. Hold-violating endpoints: 0 at all three corners | OpenSTA        | <code>report_checks -slack_max 0 -group_path_count 1000000 -endpoint_path_count 1</code> | REPRODUCED 2026-10-06 | <a href="#">sta_extra_ss_100C_1v60.log</a> section "setup violating endpoints" |

| Check                | Result                                                                                                                                                                     | Tool + version | Conditions                                                                                               | Status                | Source                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------|
| Max slew / cap at SS | 37 max-slew violating pins (worst _6342_/Y 2.004 ns vs limit 1.492 ns) and 12 max-capacitance violating pins (worst _4662_/Y 0.128 pF vs limit 0.115 pF). None at TT or FF | OpenSTA        | report_check_types - max_slew - max_capacitance - max_fanout -violators; limits from each corner liberty | REPRODUCED 2026-10-06 | <a href="#">sta_extra_ss_100C_1v60.log</a> section "drv" |

### 6.3 Baseline layout corners, for comparison only (not the reviewed layout)

Baseline variant base (no fences), build/pd/results/sky130hd/orbit\_demo/base/6\_final.\*, 7.0000 ns, same script and liberty files, run 2026-09-29 20:15.

| Corner       | Setup WNS / TNS (ns) | Hold WNS (ns) | Min period / fmax    | sep minus base min period | Status   | Source                                                                                         |
|--------------|----------------------|---------------|----------------------|---------------------------|----------|------------------------------------------------------------------------------------------------|
| tt           | +0.066 / 0.000       | +0.437        | 6.93 ns / 144.21 MHz | +0.24 ns                  | VERIFIED | <a href="#">sta_corners_baseline_7p0ns.txt</a> line 11 (= reports/pd/sky130hd/sta_corners.txt) |
| ss_100C_1v60 | -6.016 / -2240.224   | +0.893        | 13.02 ns / 76.83 MHz | +0.31 ns                  | VERIFIED | line 12                                                                                        |
| ff_n40C_1v95 | +2.256 / 0.000       | +0.280        | 4.74 ns / 210.80 MHz | +0.18 ns                  | VERIFIED | line 13                                                                                        |

The committed corner report covers only the baseline layout and its header does not name the variant; read as evidence for the sep layout it understates the SS deficit; see discrepancy register (09\_review\_notes).

### 6.4 Not modelled in any timing result

| Item                                                                    | State                                                                                                                                                                                           | Status  |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| On-chip variation, timing derates, clock uncertainty / jitter           | none in 6_final.sdc or <a href="#">pd/sta_corners.tcl</a>                                                                                                                                       | MISSING |
| I/O environment (load, drive, input transition, real I/O budget)        | not constrained; the I/O delay is the 20 % placeholder listed in 3.3                                                                                                                            | MISSING |
| RC corners                                                              | nominal OpenRCX SPEF at every corner                                                                                                                                                            | MISSING |
| Other liberty corners (e.g. ss_n40C_1v60, ff_100C_1v95, low-voltage SS) | only tt_025C_1v80 (ORFS image) and ss_100C_1v60, ff_n40C_1v95 (downloaded) exist on disk (search 2026-10-06)                                                                                    | MISSING |
| Multi-corner optimisation / closure at SS                               | ORFS optimised at TT only; SS fails at 7.2 ns                                                                                                                                                   | MISSING |
| Repo make target for corners of the sep layout                          | <a href="#">mk/pd.mk</a> pd-corners runs only build/pd/.../base; <a href="#">mk/pdsep.mk</a> has none; the 2026-10-06 run used the same commands by hand ( <a href="#">run_corners_sep.sh</a> ) | MISSING |

## 7 Max slew, max capacitance, max fanout

| Check                              | Result                                                                                                                                                                               | Tool + version              | Conditions                                                                                                          | Status                | Source                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Max slew, TT                       | 0 violations; worst slack 0.326 ns against a 1.486 ns limit                                                                                                                          | OpenSTA, OpenROAD "unknown" | TT, final STA; repair_design -cap_margin 20 -slew_margin 20 applied during the flow (SLEW_MARGIN = CAP_MARGIN = 20) | VERIFIED              | <a href="#">6_finish.rpt</a> lines 359-371, 401; <a href="#">3_4_place_resized.log</a> line 12; <a href="#">config.mk</a> lines 70-71 |
| Max capacitance, TT                | 0 violations; worst slack 0.00733 pF against a 0.02107 pF limit                                                                                                                      | same                        | same                                                                                                                | VERIFIED              | <a href="#">6_finish.rpt</a> lines 384-396, 411                                                                                       |
| Max fanout, TT                     | "0 violations" reported, but no fanout limit is defined (max_fanout_check_limit 1.0000000150474662e+30; finish_timing_drv_max_fanout_limit 0 in the metrics), so the PASS is vacuous | same                        | same                                                                                                                | VERIFIED              | <a href="#">6_finish.rpt</a> lines 374-381, 406; <a href="#">6_report.json</a> line 42                                                |
| Max fanout against a defined limit | not checked: no limit in the liberty, SDC or flow config                                                                                                                             | —                           | —                                                                                                                   | MISSING               | —                                                                                                                                     |
| Max slew / cap at SS and FF        | SS: 37 slew and 12 cap violating pins; FF: none                                                                                                                                      | OpenSTA                     | section 6.2                                                                                                         | REPRODUCED 2026-10-06 | section 6.2                                                                                                                           |

The published "max slew/cap/fanout PASS" ([results.md](#) line 17) does not name the corner and counts a fanout check that has no limit; see discrepancy register (09\_review\_notes). The metrics keys finish\_timing\_drv\_max\_slew\_limit (0.219284) and ...\_max\_cap\_limit (0.347765) in [6\\_report.json](#) are the slack-to-limit ratios printed in [6\\_finish.rpt](#) as max\_slew\_check\_slack\_limit 0.2193 and max\_capacitance\_check\_slack\_limit 0.3478, not limits.

## 8 Power

| Check                                                    | Result                                                                                                                                                                                                 | Tool + version                           | Conditions                                                                                                                                                                           | Status                | Source                                                                                                                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total power                                              | 52.1092 mW: internal 26.3633 mW, switching 25.7459 mW, leakage 0.0279 uW (2.79317e-08 W). By group: Sequential 5.96 mW (11.4 %), Combinational 43.2 mW (83.0 %), Clock 2.90 mW (5.6 %), Macro 0, Pad 0 | OpenSTA report_power, OpenROAD "unknown" | TT 25 C 1.80 V, 7.2 ns, OpenRCX SPEF; the tool's default switching activity, no VCD/SAIF annotated; the numeric default activity is not reported. A tool estimate, not a measurement | VERIFIED              | <a href="#">6_report.json</a> line 49 and power keys; <a href="#">6_finish.rpt</a> lines 549-561; <a href="#">power_default_activity.txt</a> ; <a href="#">results.md</a> lines 53-56 |
| Same query on the same database                          | 5.21e-02 W total, identical group table                                                                                                                                                                | same                                     | <a href="#">sta.tcl</a>                                                                                                                                                              | REPRODUCED 2026-10-06 | <a href="#">sta.log</a>                                                                                                                                                               |
| Activity-based power (workload VCD/SAIF), power at SS/FF | not run                                                                                                                                                                                                | —                                        | —                                                                                                                                                                                    | MISSING               | —                                                                                                                                                                                     |
| Baseline layout, for comparison                          | 47.4874 mW, same TT / default-activity conditions at 7.0 ns                                                                                                                                            | OpenSTA                                  | baseline                                                                                                                                                                             | VERIFIED              | <a href="#">results_baseline_7p0ns.md</a> line 53 (= reports/pd/sky130hd/results.md)                                                                                                  |

## 9 IR drop

| Check                      | Result                                                                                            | Tool + version                                                                                     | Conditions                                                                                                                                                                         | Status                | Source                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Static IR drop, VDD        | worst 0.4539 mV (4.53888e-04 V), average 0.0958 mV, 0.03 %; "All shapes on net VDD are connected" | OpenROAD PSM analyze_power_grid (called by ORFS final_outputs.tcl lines 39-56), OpenROAD "unknown" | supply 1.80 V (PWR_NETS_VOLTAGES), corner default, total power 5.21e-02 W from the same default-activity estimate as section 8; PSM default voltage-source placement; OpenRCX SPEF | VERIFIED              | <a href="#">6_report.log</a> lines 27-38; <a href="#">6_report.json</a> line 5; <a href="#">results.md</a> line 61 |
| Static IR drop, VSS        | worst 0.6275 mV (6.27544e-04 V), average 0.0872 mV, 0.03 %; all VSS shapes connected              | same                                                                                               | VSS 0.0 V (GND_NETS_VOLTAGES), same activity                                                                                                                                       | VERIFIED              | <a href="#">6_report.log</a> lines 39-50; <a href="#">6_report.json</a> line 9; <a href="#">results.md</a> line 62 |
| Same analysis re-run       | VDD worst 4.54e-04 V, VSS worst 6.28e-04 V, identical                                             | PSM analyze_power_grid                                                                             | <a href="#">ir.tcl</a>                                                                                                                                                             | REPRODUCED 2026-10-06 | <a href="#">ir.log</a> lines 14, 33                                                                                |
| Dynamic / vectored IR drop | not run                                                                                           | —                                                                                                  | —                                                                                                                                                                                  | MISSING               | —                                                                                                                  |

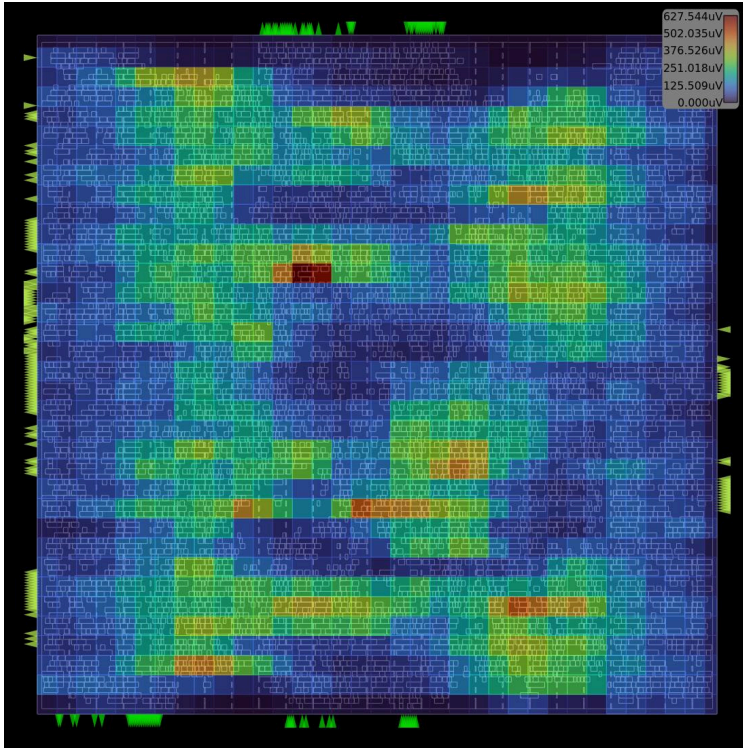


Figure 2: ORFS GUI export `final_ir_drop` (static, default activity). The export does not name the net; its legend maximum 627.544 uV equals the VSS worst-case drop, so it most likely shows VSS (ASSUMPTION). Source [final\\_ir\\_drop.webp.png](#) (= `build/pd_sep/reports/sky130hd/orbit_demo/sep/final_ir_drop.webp.png`).

## 10 Electromigration

| Check                                               | Result                                                                                                                                                                                                                                                                                                                                           | Tool + version                                 | Conditions                                                                                                          | Status                                | Source                                                                                                                                                                        |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EM in the flow                                      | NOT RUN: ORFS calls <code>analyze_power_grid</code> without <code>-enable_em</code> ; no EM text in any sep log (grep of <code>build/pd_sep/logs/...</code> and <code>make_pd_sep.out</code> , 2026-10-06)                                                                                                                                       | —                                              | —                                                                                                                   | MISSING                               | ORFS <code>final_outputs.tcl</code> lines 39-56; <a href="#">orfs_logs/</a>                                                                                                   |
| Power-grid currents only (scratch, for information) | max segment current VDD 4.05e-04 A, VSS 4.33e-04 A; no current-density limits applied, so no pass/fail                                                                                                                                                                                                                                           | PSM <code>analyze_power_grid -enable_em</code> | same default activity; per-segment reports <code>em_vdd.rpt</code> / <code>em_vss.rpt</code> stay in review scratch | REPRODUCED 2026-10-06 (currents only) | <a href="#">ir.log</a> lines 17-23, 36-42 (max current lines 20, 39)                                                                                                          |
| EM pass/fail (power grid; signal and clock nets)    | not run. <code>sky130_fd_sc_hd.tlef</code> (read by the flow, <a href="#">1_synth.log</a> line 7 ODB-0227) defines <code>DCCURRENTDENSITY</code> / <code>ACCURRENTDENSITY</code> for <code>mcon</code> , <code>via-via4</code> and <code>met1-met5</code> (15 entries); no step compared the PSM currents or signal/clock-net currents with them | —                                              | —                                                                                                                   | MISSING                               | <code>sky130_fd_sc_hd.tlef</code> (ORFS image copy; 13 layers and 25 vias, as in the ODB-0227 line) lines 91, 118-119, 135, 159-160, 178, 200-201, 218, 241-242, 260, 282-283 |

## 11 Formal property verification (fault-free RTL)

Conditions for every row: harness [orbit\\_demo\\_fv.sv](#) with an independent reference model written from [SPEC.md](#); every DUT input free in every cycle; only assumption reset in the first cycle (`clear` starts from an unconstrained state, `wrap` from a consistent arbitrary state); parameters fixed LANES = 4, T\_THROTTLE = 80, T\_STOP = 95, T\_RECOVER = 70; Yosys 0.69+154, SBY v0.69, Yices 2.7.0, bitwuzla 0.9.1; FORMAL\_TIMEOUT 1800 s. Published run 2026-09-30 00:14 UTC ([formal/summary.md](#) lines 3-4). Re-run 2026-10-06 at `3dfffbb` with a scratch BUILD. `formal/summary.md` names the RTL only as `rtl`, with no hash; see discrepancy register ([09\\_review\\_notes](#)).

### 11.1 Tasks

| Task          | Property groups                                                                                                                                  | Mode / engine / depth                      | Assertions / covers                                                                                          | Published (2026-09-30)                                    | Re-run 2026-10-06            | Status                | Source                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| thermal       | P5, P6 (+4 helpers)                                                                                                                              | prove, smtbmc yices, k-induction k = 8     | 15                                                                                                           | PASS, 0 s                                                 | PASS                         | REPRODUCED 2026-10-06 | <a href="#">formal/summary.md</a> line 28; <a href="#">logs/thermal.log</a> ; <a href="#">make_formal.log</a>             |
| dup           | P7                                                                                                                                               | prove, smtbmc yices, k = 6                 | 12                                                                                                           | PASS, 0 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 29; <a href="#">logs/dup.log</a>                                                                                     |
| handshake     | P2, P3, P4 + thermal + dup                                                                                                                       | prove, smtbmc yices, k = 8                 | 41                                                                                                           | PASS, 0 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 30; <a href="#">logs/handshake.log</a>                                                                               |
| datapath      | P1, P7, P8 (+8+8 helpers)                                                                                                                        | prove, smtbmc yices, k = 6                 | 59                                                                                                           | PASS, 3 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 31; <a href="#">logs/datapath.log</a>                                                                                |
| clear         | P5, P8 from an unconstrained start state (no reset)                                                                                              | prove, smtbmc yices, k = 3                 | 8                                                                                                            | PASS, 0 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 32; <a href="#">logs/clear.log</a>                                                                                   |
| product       | P1 lemma<br>$\text{sext32}(\text{signed8}(a) * \text{signed8}(b)) == \text{sign} * (\text{abs}(a) * \text{abs}(b))$ , all $2^{16}$ operand pairs | prove, smtbmc bitwuzla, k = 1              | 1                                                                                                            | PASS, 6 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 33; <a href="#">logs/product.log</a>                                                                                 |
| thermal_pdr   | P5, P6, no helper invariants                                                                                                                     | prove, abc pdr (unbounded)                 | 11                                                                                                           | PASS, 2 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 34; <a href="#">logs/thermal_pdr.log</a>                                                                             |
| dup_pdr       | P7                                                                                                                                               | prove, abc pdr                             | 12                                                                                                           | PASS, 2 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 35; <a href="#">logs/dup_pdr.log</a>                                                                                 |
| handshake_pdr | P2, P3, P4, no helpers                                                                                                                           | prove, abc pdr                             | 13                                                                                                           | PASS, 7 s                                                 | PASS                         | REPRODUCED 2026-10-06 | line 36; <a href="#">logs/handshake_pdr.log</a>                                                                           |
| live          | P2_live_delivered_s_eventually<br>!f_watch under fairness_s_eventually<br>out_ready                                                              | live, aiger suprove (liveness-to-safety)   | 1                                                                                                            | PASS, 25 s; suprove gives no per-property status or trace | PASS, 17 s                   | REPRODUCED 2026-10-06 | line 37; <a href="#">logs/live.log</a>                                                                                    |
| cover         | C01-C14 from reset                                                                                                                               | cover, smtbmc yices, depth 24              | 14 covers, all reached at step 3-6                                                                           | PASS                                                      | PASS, same steps             | REPRODUCED 2026-10-06 | lines 38, 44-59; <a href="#">logs/cover.log</a>                                                                           |
| wrap          | C20-C22 INT32 wraparound from a consistent arbitrary state                                                                                       | cover, smtbmc yices, depth 8, cover_assert | 3 covers reached at step 2; 88 assertions UNKNOWN (checked only along the 3 traces, not proven by this task) | PASS                                                      | PASS                         | REPRODUCED 2026-10-06 | lines 39, 60-62, 313-404; <a href="#">logs/wrap.log</a>                                                                   |
| datapath_pdr  | P1, P8, no helpers (make formal-extra only)                                                                                                      | prove, abc pdr                             | 31                                                                                                           | PASS, 837 s                                               | PASS, 941 s (15 m 41 s wall) | REPRODUCED 2026-10-06 | line 40; <a href="#">logs/datapath_pdr.log</a> ; <a href="#">make_formal-extra.log</a> ; <a href="#">summary_extra.md</a> |

Re-run totals: make formal "[formal] all 12 tasks passed", 46.6 s wall, exit 0 ([make\\_formal.log](#)); per-property status P1-P8 and cover steps identical to the published summary ([formal\\_summary.md](#) lines 13-62).

## 11.2 Per-property result

| Property           | Meaning (short)                                                                                                                                                                                                   | Result                                                                                                                                                                                          | Status                | Source                                          |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------|
| P1                 | every out_fire delivers the reference result per lane, INT32 wraparound                                                                                                                                           | PROVEN unbounded (datapath k = 6, product, datapath_pdr)                                                                                                                                        | REPRODUCED 2026-10-06 | <a href="#">formal/summary.md</a> line 15       |
| P2                 | every accepted in_last gives exactly one out_fire, in order                                                                                                                                                       | PROVEN unbounded (handshake, handshake_pdr, live)                                                                                                                                               | REPRODUCED 2026-10-06 | line 16                                         |
| P3                 | out_valid && !out_ready: held and stable                                                                                                                                                                          | PROVEN unbounded                                                                                                                                                                                | REPRODUCED 2026-10-06 | line 17                                         |
| P4                 | in_ready low in STOP/3, fault, clear_fault, buffer full && !out_ready                                                                                                                                             | PROVEN unbounded; the code-3 branch is vacuous because voted code 3 is unreachable fault-free (P5_never_code3) and mutant m27 survives (section 14); see discrepancy register (09_review_notes) | REPRODUCED 2026-10-06 | line 18                                         |
| P5                 | voted thermal state follows the SPEC table; reset gives STOP                                                                                                                                                      | PROVEN unbounded (P5_shutdown_req vacuous for code 3, as P4)                                                                                                                                    | REPRODUCED 2026-10-06 | line 19                                         |
| P6                 | throttle: first THROTTLE cycle admits, then alternate                                                                                                                                                             | PROVEN unbounded                                                                                                                                                                                | REPRODUCED 2026-10-06 | line 20                                         |
| P7                 | fault-free: fault and therm_repair never rise, copies equal                                                                                                                                                       | PROVEN unbounded                                                                                                                                                                                | REPRODUCED 2026-10-06 | line 21                                         |
| P8                 | clear_fault gives fault = 0, out_valid = 0, storage zero, next beat sums from 0 (also from a latched fault)                                                                                                       | PROVEN unbounded                                                                                                                                                                                | REPRODUCED 2026-10-06 | line 22                                         |
| Induction depth    | "closes at k = 2 for thermal, handshake and datapath and at k = 1 for dup (measured 2026-09-29)"; the logs are consistent (third / second "Trying induction" step succeeds) but the measurement itself has no log | —                                                                                                                                                                                               | CLAIM (unverified)    | <a href="#">formal/summary.md</a> lines 504-508 |
| Development claims | "a first attempt ran for over 10 minutes at step 4"; the clear task "added after an independent review found ..."                                                                                                 | no log                                                                                                                                                                                          | CLAIM (unverified)    | lines 499-500, 516-517                          |

SPEC section 6 says phase is "forced to 0 whenever the voted state is not THROTTLE"; the RTL clears it at the next edge. The formal summary documents the difference (lines 525-532) and P6 proves admission is unaffected; see discrepancy register (09\_review\_notes).

## 11.3 Vacuity check (19 one-line RTL mutants that the proofs must fail)

Method: make formal-vacuity applies one sed edit to a generated RTL copy (never rtl/) and runs the task that should catch it; CAUGHT = sby FAIL with a counterexample from reset on a property with the expected prefix; ERROR, UNKNOWN or timeout (FORMAL\_VAC\_TIMEOUT 900 s) count as MISSED ([formal/vacuity.md](#) lines 3-7).

| Result                                            | Tool + version                                    | Conditions                                              | Status   | Source                                                                                   |
|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|----------|------------------------------------------------------------------------------------------|
| 19 / 19 CAUGHT (published 2026-09-30 00:13:00:14) | SBY v0.69, smtbmc yices / abc pdr / aiger suprove | 14 mutants on the k-induction tasks, 5 on the pdr tasks | VERIFIED | <a href="#">formal/vacuity.md</a> lines 11-29; <a href="#">formal/logs/vacuity_*.log</a> |

| Result                                                 | Tool + version | Conditions                       | Status                | Source                                                                      |
|--------------------------------------------------------|----------------|----------------------------------|-----------------------|-----------------------------------------------------------------------------|
| 19 / 19 CAUGHT, "[vacuity] all mutants caught", 31.4 s | same           | re-run at 3dffffb, scratch BUILD | REPRODUCED 2026-10-06 | <a href="#">formal_vacuity.md</a> ; <a href="#">make_formal-vacuity.log</a> |
| 19 / 19 CAUGHT, 31.0 s (second, independent re-run)    | same           | separate scratch BUILD           | REPRODUCED 2026-10-06 | <a href="#">make_formal-vacuity_second_run.log</a>                          |

The verdicts are stable; the failing property and step reported for two abc-pdr mutants are not:

| Mutant              | Published                                                       | Re-run A (package formal_rerun)      | Re-run B (second run)                                           |
|---------------------|-----------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------|
| zext_product_pdr    | P1_out_valid_data_lane1<br>P8_fresh_sum_from_zero_lane1, step 4 | P8_fresh_sum_from_zero_lane2, step 6 | P1_out_valid_data_lane3<br>P8_fresh_sum_from_zero_lane3, step 4 |
| clear_keeps_acc_pdr | P8_clear_zero_acc_a_lane0<br>P8_clear_zero_acc_b_lane0, step 5  | same properties, step 8              | same properties, step 7                                         |

The other 17 rows (property names and steps) are identical in all three runs. The "step 0" entries for no\_backpressure and no\_backpressure\_pdr cannot be checked from the trimmed published logs; see discrepancy register (09\_review\_notes).

## 11.4 Formal limits

Fault-free RTL only (fault detection is section 12); RTL only, no property check on any netlist; single clock; parameters fixed; covers from reset searched to depth 24; the harness depends on Yosys hierconn. Source: [formal/summary.md](#) lines 534-549. Status of any netlist-level property check: MISSING.

## 12 Fault injection

### 12.1 Formal fault injection on generated RTL copies

Conditions: [fault/gen\\_fault\\_copies.py](#) writes one RTL copy per scenario with one storage element made upsettable ( $q \leq q_{\text{next}} \wedge \text{fi\_mask}, \text{fi\_bit} = \text{\$anyconst}, \text{fi\_req} = \text{\$anyseq}$ ); production RTL is the fault-free twin in [fi\\_harness.sv](#); at most one upset per trace, any cycle including reset cycles, any lane, any bit; protected scenarios: prove with k-induction  $k = 4$  (FAULT\_PROVE\_DEPTH), cover depth 12; negative scenarios: BMC depth 12; engine smtbmc yices; Yosys 0.69+154, SBY v0.69, Yices 2.7.0. Injection is on RTL copies, not on any netlist. Published run 2026-09-30 00:41-00:44; re-run 2026-10-06 with `make BUILD=<scratch> FAULT_PUBLISH= fault` (reports/fault not written), "[fault] all checks passed", 3 m 17 s.

| Scenario                                                                                                       | Injected element                                              | Result                                                                                                                  | Status                | Source                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| acc_a, acc_b                                                                                                   | u_acc_a, u_acc_b, any lane/bit/cycle                          | prove PASS, 18 assertions, induction closed; cover 8/8                                                                  | REPRODUCED 2026-10-06 | <a href="#">fault/summary.md</a> lines 45-46; <a href="#">fault/logs/acc_a_prove.log</a> ; <a href="#">formal_results.md</a> |
| res_a, res_b                                                                                                   | u_res_a (drives out_data), u_res_b                            | prove PASS, 18 assertions, induction closed; cover 8/8                                                                  | REPRODUCED 2026-10-06 | lines 47-48; <a href="#">fault/logs/</a>                                                                                     |
| therm_c0, therm_c1, therm_c2                                                                                   | u_thermal.u_copy0/1/2, any bit/cycle                          | prove PASS, 16 assertions, induction closed; cover 8/8 (incl. C_upset_copy_code3, C_upset_in_throttle, C_upset_in_stop) | REPRODUCED 2026-10-06 | lines 49-51                                                                                                                  |
| neg_out_valid_q (unprotected)                                                                                  | out_valid_q                                                   | cover 5/5; fire FAIL as expected (D_out_fire_matches_ref, step 1); loss FAIL as expected (D_no_silent_loss, step 3)     | REPRODUCED 2026-10-06 | line 52; <a href="#">neg_out_valid_q_fire.log</a> , <a href="#">neg_out_valid_q_loss.log</a>                                 |
| neg_product (unprotected)                                                                                      | one-cycle transient on the shared prod32 of one lane          | cover 2/2; fire FAIL as expected (D_out_fire_matches_ref, step 3); wrong out_data transferred with no fault (SDC)       | REPRODUCED 2026-10-06 | line 53; <a href="#">neg_product_fire.log</a> ; <a href="#">neg_product.diff</a>                                             |
| Negative control no_compare (assign mismatch = 1'b0;)                                                          | acc_a, res_a on the mutant                                    | 4 / 4 FAIL as expected (L_stop_in_upset_cycle step 1; D_out_fire_matches_ref step 3)                                    | REPRODUCED 2026-10-06 | lines 143-163; <a href="#">negctl_no_compare.diff</a> ; <a href="#">negctl_formal_results.md</a>                             |
| Totals                                                                                                         | 19 scenario tasks, 0 unexpected; 4 negctl tasks, 0 unexpected | —                                                                                                                       | REPRODUCED 2026-10-06 | <a href="#">make_fault.log</a> ; <a href="#">summary.md</a> (re-run)                                                         |
| Hand-run BMC without helpers (about step 8 / frame 10 after about 10 min) and 24-step BMC with helpers (7 min) | no log                                                        | —                                                                                                                       | CLAIM (unverified)    | <a href="#">fault/summary.md</a> lines 57-62                                                                                 |

### 12.2 RTL SEU simulation campaign

Conditions: [tb\\_seu\\_campaign.v](#), Icarus Verilog 14.0 (devel); two copies of the production RTL behind identical hosts on one random workload; seed=1 tpb=12 tpb\_unprot=200 controls=20 prefix=8..127 post=64 drain=24+8; one bit flipped by hierarchical assignment at the falling edge; zero-delay RTL, 10 ns sim clock; host reaction clear\_fault 2 cycles after fault. Re-run 2026-10-06: [trials.tsv](#) byte-identical to [build/fault/campaign/trials.tsv](#), report byte-identical to the published [campaign.md](#).

| Storage group                 | Result (trials: outcomes)                                                                                                                                                                                       | Status                | Source                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| All                           | 6836 trials, 0 bench errors; 20 controls all MASKED; 6816 injected; all 521 bits, each at least 12 times; 9 / 9 campaign checks PASS                                                                            | REPRODUCED 2026-10-06 | <a href="#">fault/campaign.md</a> lines 12, 36-46; <a href="#">campaign_report.md</a> ; per-bit table <a href="#">campaign_per_bit.tsv</a> |
| Accumulator pairs (256 bits)  | 3072: MASKED 3, DETECTED 3069, escapes 0                                                                                                                                                                        | REPRODUCED 2026-10-06 | <a href="#">campaign.md</a> line 5                                                                                                         |
| Result pairs (256 bits)       | 3072: MASKED 4, DETECTED 3067, TIMING 1, escapes 0                                                                                                                                                              | REPRODUCED 2026-10-06 | line 6                                                                                                                                     |
| Detection latency (acc + res) | all 6136 detected trials: fault = 1 one cycle after the upset cycle; in_ready and out_valid already 0 at the end of the upset cycle                                                                             | REPRODUCED 2026-10-06 | line 14                                                                                                                                    |
| 8 acc/res upsets not DETECTED | all had clear_fault in the upset cycle (both copies zeroed); 7 MASKED, 1 TIMING (trial 3836 g_lane[1].u_lane.u_res_a.q[30])                                                                                     | REPRODUCED 2026-10-06 | lines 16-17                                                                                                                                |
| Thermal triple (6 bits)       | 72: REPAIRED 72, escapes 0                                                                                                                                                                                      | REPRODUCED 2026-10-06 | line 7                                                                                                                                     |
| Throttle phase (unprotected)  | 200: MASKED 156, TIMING 44 (all in THROTTLE). Escapes column 0 because TIMING is not counted as an escape although SPEC section 8 lists phase as an expected escape; see discrepancy register (09_review_notes) | REPRODUCED 2026-10-06 | lines 8, 25-27                                                                                                                             |
| out_valid_q (unprotected)     | 200: LOST 38, EXTRA 162 (129 duplicate, 33 stale), escapes 200 (100 %)                                                                                                                                          | REPRODUCED 2026-10-06 | lines 9, 23-24, 30                                                                                                                         |
| fault_q 0 -> 1 (unprotected)  | 200: DETECTED 200, a false fault stop with no data error                                                                                                                                                        | REPRODUCED 2026-10-06 | lines 10, 28                                                                                                                               |

| Storage group                           | Result (trials: outcomes)                                                                          | Status                | Source                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------|
| fault_q 1 -> 0 (SPEC section 8 escape)  | not measured: needs two faults, the single-upset campaign cannot reach fault_q = 1 first           | MISSING               | <a href="#">fault/summary.md</a> lines 254-256                                            |
| Negative control campaign on no_compare | 535 trials, 0 errors; acc 256: MASKED 196, SDC 60; res 256: MASKED 230, SDC 26; escape checks PASS | REPRODUCED 2026-10-06 | <a href="#">fault/summary.md</a> lines 230-239; <a href="#">negctl_campaign_report.md</a> |

### 12.3 Post-route gate-level fault campaign (sep netlist)

| Check                                     | Result                                                                                                                                                                                                                                                                     | Tool + version                                                                                                          | Conditions                                                                                                                                                                                                                                                  | Status                | Source                                                                                                                                 |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| GLS lockstep with injection               | "GLS PASS: 30000 cycles, 0 mismatches"; 702 duplicated-copy flips (702 stopped), 234 thermal-copy flips (234 repaired); 518 of 518 redundant bits hit; coverage: 8729 beats, 1037 results, 6 resets, 733 clears; NORMAL 14075 / THROTTLE 1737 / STOP 14182 cycles, code3 0 | Icarus Verilog (version not logged in 2026-09-29 run); cell models from cells.lib (tt_025C_1v80) via Yosys read_liberty | <a href="#">tb_gls.v</a> : RTL and routed sep 6_final.v in lockstep, zero-delay, seed 1, one flip per 32 cycles at the falling edge (list in <a href="#">gl_inject.vh</a> ); checks stop/repair behaviour and lockstep outputs, no golden-result scoreboard | VERIFIED              | <a href="#">gls/gls.log</a> lines 1-5 (= build/pd_sep/gls/sep/gls.log = reports/pdsep/gls.log); <a href="#">pdsep.mk</a> lines 127-137 |
| Same campaign re-run                      | identical output, 2 m 18 s                                                                                                                                                                                                                                                 | Icarus Verilog 14.0 (devel) s20260301-500-g2e81fccb-dirty                                                               | same                                                                                                                                                                                                                                                        | REPRODUCED 2026-10-06 | <a href="#">gls_rerun.log</a>                                                                                                          |
| Netlist-mutant control on the sep netlist | instance_4098_sky130_fd_sc_hd_and2_1 -> sky130_fd_sc_hd_or2_1 (#0 of 184): first MISMATCH at cycle 7, "GLS FAIL: 1148 errors" (the harness observes the netlist)                                                                                                           | Icarus 14.0                                                                                                             | 3000 cycles, seed 1. The log does not name the netlist; "#0 of 184" matches the 184 and2_1 instances of the sep 6_final.v (the base netlist has 189)                                                                                                        | REPRODUCED 2026-10-06 | <a href="#">gls_mutant.log</a>                                                                                                         |
| Directed bench on the sep netlist         | see <a href="#">05_simulation</a> (S8): TB_ORBIT_DEMO PASS, summary identical to RTL                                                                                                                                                                                       | Icarus 14.0                                                                                                             | zero-delay                                                                                                                                                                                                                                                  | REPRODUCED 2026-10-06 | <a href="#">gls_sky130hd_sep_directed.log</a>                                                                                          |

### 12.4 Not injected

| Item                                                                                    | Status                                                          | Source                                                   |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Comparators (lane_mismatch, mismatch), voter, clock, reset, sensor and handshake inputs | MISSING (listed in SPEC section 8, not injected in either flow) | <a href="#">fault/summary.md</a> lines 243-246           |
| Multi-bit upsets, common-mode upsets of both copies                                     | MISSING                                                         | same                                                     |
| fault_q 1 -> 0 (needs two faults); thermal voted code 3 (needs two copies upset)        | MISSING                                                         | lines 254-256; <a href="#">mutation/summary.md</a> (m27) |
| Shared-product transient in the simulation campaign (only formal neg_product)           | MISSING                                                         | campaign injects stored bits only                        |
| Fault injection with SDF timing (SET width, latching window)                            | MISSING                                                         | all GLS zero-delay                                       |
| Multi-seed campaign statistics, confidence intervals                                    | MISSING (seed 1 only)                                           | <a href="#">fault/summary.md</a> lines 251-253           |
| Radiation test data, cross-sections, FIT                                                | MISSING ("Nothing here is a radiation test")                    | line 257                                                 |

## 13 Synthesis checks

| Check                                                                 | Result                                                                                                                                                                                                                                                                                                                                                                                                          | Tool + version                                       | Conditions                                   | Status                | Source                                                                                         |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------|
| keep_hierarchy in the ORFS (reviewed) synthesis                       | hierarchy kept: 19 orbit_keep_reg submodules (16 x W=32, 3 x W=2 RESET_VAL=2'10); 521 sky130_fd_sc_hd_dfxtpt_1; 5130 cells, 46866.1984 um^2. Kept by (* keep_hierarchy *) in <a href="#">orbit_keep_reg.v</a> line 14 and SYNTH_KEEP_MODULES = *orbit_keep_reg* ( <a href="#">config.mk</a> line 30). Log warning Selection "*orbit_keep_reg*\\$" did not match any module (the second, slang-frontend pattern) | Yosys 0.68+post (ORFS)                               | ABC speed script, -D 7.2, liberty TT         | VERIFIED              | <a href="#">synth_stat.txt</a> lines 152-154, 265-269; <a href="#">1_2_yosys.log</a> lines 1-9 |
| ORFS synthesis structural check                                       | CHECK pass "Found and reported 0 problems" on both orbit_keep_reg variants and orbit_demo                                                                                                                                                                                                                                                                                                                       | Yosys 0.68+post                                      | —                                            | VERIFIED              | <a href="#">synth_check.txt</a>                                                                |
| keep negative control, ORFS (baseline platform, synthesis stage only) | cfgkeep (RTL attribute removed, SYNTH_KEEP_MODULES set): PASS 19/19, 521 FF; nokeep (both removed): FAIL 17/19, 517 FF, thermal u_copy1/u_copy2 0 cells                                                                                                                                                                                                                                                         | Yosys 0.68+post, <a href="#">pd/audit_storage.py</a> | baseline pd/sky130hd config, not the sep run | VERIFIED              | <a href="#">negctl_cfgkeep_audit.txt</a> ; <a href="#">negctl_nokeep_audit.txt</a>             |
| keep negative control, generic flow                                   | nokeep: 517 FF, 4865 cells, thermal copies MERGED, audit FAIL 8/9; regkeep ((* keep *) on q only): same, does not prevent the merge; NEGATIVE_CONTROL PASS                                                                                                                                                                                                                                                      | Yosys 0.69+154                                       | <a href="#">synth/synth_generic.py</a>       | REPRODUCED 2026-10-06 | <a href="#">negative_control.txt</a> (re-run byte-identical, <a href="#">make_synth.log</a> )  |

| Check                                                                         | Result                                                                                                                                                                                                                                                                            | Tool + version                                                                                       | Conditions                                                             | Status                | Source                                                                                                                |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Storage audit, routed sep 6_final.v (published)                               | RESULT: PASS (18/18 checks passed); 6250 leaf cells; 521 dfxtp_1; 9 redundant groups OK; 521 distinct Q nets; 51 clock-leaf nets. Produced by the older pd/audit_storage.py (report 2026-09-29 22:30; script changed in 798edc7, originally committed 23:47, adding a 19th check) | pd/audit_storage.py (pre-798edc7), liberty build/pd_sep/platform/cells.lib                           | —                                                                      | VERIFIED              | <a href="#">storage_audit.txt</a> lines 3, 36                                                                         |
| Storage audit, same netlist, current script                                   | RESULT: PASS (19/19); added check "clock pins on the clock tree: 0 flip-flops whose clock net does not trace back to port 'clk'" PASS; all other lines identical                                                                                                                  | <a href="#">pd/audit_storage.py</a> at HEAD                                                          | same netlist and liberty                                               | REPRODUCED 2026-10-06 | <a href="#">storage_audit_current_script.txt</a> lines 35, 37                                                         |
| Storage audit, generic netlist                                                | STORAGE_AUDIT PASS: 9/9 groups, 521 FF bits, 3 unprotected (fault_q, out_valid_q, u_thermal_phase); flat view identical; self-test: 12 defect classes all detected                                                                                                                | Yosys 0.69+154, <a href="#">scripts/storage_audit.py</a>                                             | generic netlist build/synth/orbit_demo_synth.json                      | REPRODUCED 2026-10-06 | <a href="#">synth/storage_audit.txt</a> ; <a href="#">storage_audit_flat.txt</a> ; <a href="#">audit_selftest.log</a> |
| Generic synthesis structural checks                                           | SYNTH_GENERIC PASS: 4899 cells = 4373 generic logic + 521 FF + 5 \$scopeinfo; no latches; no warnings                                                                                                                                                                             | Yosys 0.69+154                                                                                       | synth -flatten with kept instances                                     | REPRODUCED 2026-10-06 | <a href="#">synth_counts.txt</a> (re-run byte-identical)                                                              |
| Equivalence RTL vs generic netlist                                            | EQUIV PASS: 47 register ports cut and matched by name; tasks lane0-lane3, ctrl PASS; mutants mut_throttle (T_THROTTLE 80 -> 81) and mut_zext FAIL as required                                                                                                                     | SymbiYosys, smtbmc bitwuzla, BMC depth 1 per partition (register-correspondence combinational check) | gold RTL vs build/synth/orbit_demo_synth.v (generic cells, not sky130) | REPRODUCED 2026-10-06 | <a href="#">equiv_result.txt</a> ; re-run (44 s vs 40 s wall)                                                         |
| GLS of the generic netlist                                                    | TB_ORBIT_DEMO PASS, summary identical to the RTL run (14 scenarios, 133601 cycles, 0 errors)                                                                                                                                                                                      | Icarus 14.0, Yosys simcells/simlib                                                                   | zero-delay, seed 1                                                     | REPRODUCED 2026-10-06 | <a href="#">gls_result.txt</a> ; <a href="#">make_synth.log</a>                                                       |
| make synth overall                                                            | "SYNTH PASS: generic synthesis + checks, storage audit (+ self-test), keep_hierarchy negative control, equivalence to RTL, gate-level simulation", 2 m 0.2 s                                                                                                                      | Yosys 0.69+154                                                                                       | scratch BUILD                                                          | REPRODUCED 2026-10-06 | <a href="#">make_synth.log</a>                                                                                        |
| RTL vs sky130-mapped netlist (1_2_yosys.v / 1_synth_lec.v) formal equivalence | not run; only zero-delay GLS links RTL to the sky130 netlists (12.3)                                                                                                                                                                                                              | —                                                                                                    | —                                                                      | MISSING               | —                                                                                                                     |
| Brief p.4 "4,356 generic logic cells"                                         | not reproduced: 4373 native, 4384 YoWASP Yosys 0.69; no script given                                                                                                                                                                                                              | —                                                                                                    | —                                                                      | CLAIM (unverified)    | <a href="#">synth/summary.md</a> lines 12, 21; <a href="#">yowasp_compare.txt</a>                                     |

Two different storage audits exist (generic [scripts/storage\\_audit.py](#), "9/9 groups"; sky130 [pd/audit\\_storage.py](#), "18/18" or "19/19 checks"), and the published sep reports (results.md line 24, checks.json, summary.md) still quote the superseded 18/18; see discrepancy register (09\_review\_notes).

## 14 Mutation testing of the verification suite (context for formal and fault)

| Check          | Result                                                                                                                                                                                                                                                                                                                                                                             | Tool + version                                                                                   | Conditions                                                                        | Status   | Source                                           |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|--------------------------------------------------|
| Mutation score | 35/40 valid mutants killed (87.5 %); 35/39 (89.7 %) excluding m21, argued equivalent                                                                                                                                                                                                                                                                                               | scripts/mutation_test.py; targets lint, sim, formal-quick, fault-quick, synth                    | 40 RTL mutants, baseline m00 passes all targets; generated 2026-09-30; not re-run | VERIFIED | <a href="#">mutation/summary.md</a> lines 5-52   |
| Survivors      | m07 result_from_b, m20 out_valid_no_fault, m22 fault_not_sticky, m27 shutdown_only_stop (real test gaps), m21 out_fire_unmasked (equivalent); all also pass the full formal set. Missing checks named: fault -> !out_valid && !in_ready and fault stickiness under a fault_q upset; out_data equal to reference while out_valid low; therm_state == 3 -> shutdown_req && !in_ready | evidence bench <a href="#">tb_mutation_gaps.v</a> FAILS on each survivor, PASS on unmodified RTL | —                                                                                 | VERIFIED | <a href="#">mutation/summary.md</a> lines 72-140 |

## 15 ECC standalone block (not in the layout)

[rtl/ecc/orbit\\_secdded72.v](#) (Hsiao (72,64) SECDED encoder/decoder) and [rtl/ecc/orbit\\_ecc\\_bank.v](#) (16 rows x 2 interleaved codewords, scrubber, error log) are not instantiated by orbit\_demo, are not in \$(RTL), and were not read by the ORFS synthesis of the reviewed layout ([1\\_1\\_yosys\\_canonicalize.log](#) lines 7-11 read only the four orbit\_demo RTL files; [mk/ecc.mk](#) line 3 "NOT part of orbit\_demo"). Physical checks (timing, area, power, DRC, LVS, IR, EM) of ECC: N/A for this layout (block not placed); MISSING if ECC is to be integrated.

Conditions: published run [ecc/results.txt](#) (= reports/ecc/results.txt); full make BUILD=<scratch> ecc re-run 2026-10-06 at 3dfffbb: "ECC: PASS", exit 0, 20 m 41 s; summary identical to the published results except run times. Tools Yosys 0.69+154, SBY, smtbmc bitwuzla/yices, Icarus 14.0, Verilator 5.053.

| Check | Result                                 | Tool                         | Conditions                  | Status                | Source                                                               |
|-------|----------------------------------------|------------------------------|-----------------------------|-----------------------|----------------------------------------------------------------------|
| Lint  | Verilator -Wall and Icarus -Wall clean | Verilator 5.053, Icarus 14.0 | --top-module orbit_ecc_bank | REPRODUCED 2026-10-06 | <a href="#">ecc/results.txt</a> line 14; <a href="#">summary.txt</a> |

| Check                                                          | Result                                                                                                                                                                                              | Tool                                               | Conditions                                                                                                                                                            | Status                          | Source                                                                                                 |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------|
| H matrix                                                       | Hsiao construction: 72 distinct odd-weight columns, 26 ones per data row                                                                                                                            | <a href="#">rtl/ecc/gen_hsiao72.py</a><br>-- check | —                                                                                                                                                                     | REPRODUCED<br>2026-10-06        | results.txt line 13; <a href="#">hsiao.log</a>                                                         |
| Codec simulation                                               | seed 1: 20003 no-error words; 21600 single-bit errors corrected (300 words x 72); 30672 double-bit errors detected (12 x 2556 pairs); 19189 triple-bit: 8417 UE, 10772 miscorrected; errors 0; PASS | Icarus 14.0                                        | reference H built from the construction rule                                                                                                                          | REPRODUCED<br>2026-10-06        | <a href="#">sim_codec_seed1.log</a>                                                                    |
| Bank simulation                                                | seeds 1 and 2, 15098 cycles each (15000 random + directed), errors 0; seed 1: 3462 writes / 4971 reads, 270 CE / 14 UE reads, 1075 scrub repairs                                                    | Icarus 14.0                                        | DEPTH 16, INTERLEAVE 2, CNT_W 16 plus a CNT_W 3 copy; upsets by hierarchical writes, never a third error per codeword                                                 | REPRODUCED<br>2026-10-06        | <a href="#">sim_bank_seed1.log</a> , <a href="#">sim_bank_seed2.log</a>                                |
| Simulation negative controls                                   | 7 / 7 mutants killed (bank_cnt_wrap, bank_log_prio, bank_no_interleave, bank_no_writeback, bank_read_way_swap, bank_scrub_on_read, codec_even_col)                                                  | Icarus 14.0                                        | exact text substitution, bank bench 3000 cycles                                                                                                                       | REPRODUCED<br>2026-10-06        | results.txt lines 18-24                                                                                |
| Codec proof                                                    | PASS: no error unchanged; any single error corrected (ce=1); any double error ue=1, no flip; encoder linear                                                                                         | SBY prove depth 1, smtbmc bitwuzla                 | combinational, all inputs free (complete)                                                                                                                             | REPRODUCED<br>2026-10-06        | results.txt line 8; <a href="#">formal_codec_orbit_secdded72_codec.log</a>                             |
| Bank proofs clean / sec / ded                                  | PASS by k-induction (41 s / 135 s / 89 s published)                                                                                                                                                 | SBY prove depth 2, smtbmc bitwuzla                 | DEPTH 16, INTERLEAVE 2, CNT_W 3; decoder replaced by an abstraction constrained to the codec contract; at most one upset event per edge (1 bit or 2 adjacent columns) | REPRODUCED<br>2026-10-06        | results.txt lines 1, 4, 7; <a href="#">formal_bank_*.log</a>                                           |
| Covers                                                         | cover: c_scrub_two step 2, c_ce_read_adj 3, c_saturate 6, c_bound_tight 17; cover_ded: c_ue_read step 4                                                                                             | SBY cover depth 24, smtbmc yices                   | —                                                                                                                                                                     | REPRODUCED<br>2026-10-06        | results.txt lines 2-3                                                                                  |
| End-to-end with the real decoder                               | e2e_clean PASS (BMC 5 steps), e2e_upset PASS (BMC 4 steps); bounded only; U4 scrub bound not reached at this depth                                                                                  | SBY bmc, bitwuzla                                  | DEPTH 2 (chparam)                                                                                                                                                     | REPRODUCED<br>2026-10-06        | results.txt lines 5-6                                                                                  |
| Unbounded end-to-end statement (real encoder + decoder + bank) | rests on a written composition argument, not machine-checked                                                                                                                                        | —                                                  | —                                                                                                                                                                     | CLAIM<br>(unverified)           | <a href="#">rtl/ecc/README.md</a> lines 186-206                                                        |
| Formal negative controls                                       | 4 / 4 FAIL as expected (il1_neg, codec_even_col, bank_no_writeback, bank_read_way_swap)                                                                                                             | SBY                                                | —                                                                                                                                                                     | REPRODUCED<br>2026-10-06        | results.txt lines 9-12                                                                                 |
| sec_bmc positive control, published log                        | incomplete run of an older harness: steps 0-3 without failure, terminated at 600 s during step 4; neither PASS nor FAIL; not listed in results.txt                                                  | SBY bmc depth 6, bitwuzla                          | DEPTH 16, INTERLEAVE 2, CNT_W 3; harness before commit 826bd11                                                                                                        | VERIFIED (as an incomplete run) | <a href="#">formal_sanity_sec_bmc_production_timeout.log</a>                                           |
| sec_bmc positive control, re-run with the current harness      | PASS at depth 6 (steps 0-5), 2672 s                                                                                                                                                                 | SBY bmc depth 6, bitwuzla                          | same parameters, harness at 3dffffb                                                                                                                                   | REPRODUCED<br>2026-10-06        | <a href="#">sanity_sec_bmc_result.txt</a> ; <a href="#">sanity_sec_bmc_production_pass_trimmed.log</a> |
| Generic synthesis                                              | enc 154 cells, 0 FF, depth 6; dec 431 cells, 0 FF, depth 18; bank 10690 cells, 2414 FF (expected 2414), depth 48; no latches                                                                        | Yosys 0.69+154                                     | generic cells, no liberty                                                                                                                                             | REPRODUCED<br>2026-10-06        | results.txt lines 25-27; <a href="#">ecc/synth_stat.txt</a>                                            |

The stale sec\_bmc timeout log and the orphan [sim\\_bank\\_quick.log](#) (= reports/ecc/logs/sim\_bank\_quick.log) are flagged; see discrepancy register (09\_review\_notes).

## 16 Not done / missing

| Item                                              | Why it matters here                                                                                                                                                                                | What would be needed                                                          | Status  |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------|
| Sign-off DRC (FEOL, density, latch-up), Magic DRC | KLayout ORFS deck runs BEOL + OFFGRID only; no fill                                                                                                                                                | Magic DRC or the open_pdk KLayout sign-off deck on a filled GDS               | MISSING |
| Metal fill and density checks                     | no fill inserted; sky130_fd_sc_hd.tlef (read by the flow, 1_synth.log ODB-0227) defines MAXIMUMDENSITY 70 / DENSITYCHECKWINDOW 700 700 (met1-met4), but no step compared the metal density with it | fill generation, then a density check against these values or a sign-off deck | MISSING |
| Independent LVS                                   | KLayout LVS reference comes from the same 6_final.odb                                                                                                                                              | Magic extraction + Netgen against 6_final.v or the synthesized netlist        | MISSING |
| Antenna check outside OpenROAD                    | only the OpenROAD ANT checker ran                                                                                                                                                                  | Magic / KLayout antenna rules                                                 | MISSING |
| RTL vs sky130 netlist formal equivalence          | kepler-formal covers synth -> final only; RTL -> generic proven, RTL -> sky130 only simulated                                                                                                      | eqy or equiv_cut.yf flow with sky130 cell models on 1_2_yosys.v / 6_final.v   | MISSING |

| Item                                                                                                           | Why it matters here                                                                                                                                                                                                                   | What would be needed                                                             | Status                                                                                     |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| kepler-formal, OpenROAD, ORFS, PDK versions                                                                    | not recorded in any log                                                                                                                                                                                                               | log image digest, ORFS/OpenROAD commit, open_pdk version per run                 | MISSING                                                                                    |
| RC corners, OCV / derates, clock uncertainty, I/O budgets                                                      | timing uses nominal RC and placeholder I/O delays                                                                                                                                                                                     | min/max RCX rules, uncertainty/derate spec, interface budget                     | MISSING                                                                                    |
| Additional liberty corners; timing closure at SS                                                               | SS fails by 6.131 ns with 516 endpoints and 37/12 slew/cap violations                                                                                                                                                                 | more corners; MCMC optimisation, or a decision on operating corner and period    | MISSING                                                                                    |
| Demonstrator speed / PVT requirement                                                                           | no requirement to judge corner results against                                                                                                                                                                                        | stated target frequency and PVT range                                            | MISSING                                                                                    |
| Post-layout SPICE; SDF-annotated GLS                                                                           | all GLS zero-delay; no transistor-level simulation                                                                                                                                                                                    | SDF from OpenSTA with 6_final.spf; timing GLS; SPICE of critical paths if needed | MISSING                                                                                    |
| Activity-based and corner power; dynamic IR                                                                    | power and IR use default activity at TT                                                                                                                                                                                               | VCD/SAIF from a workload; read in OpenSTA; vectored PSM                          | MISSING                                                                                    |
| Electromigration                                                                                               | not run; sky130_fd_sc_hd.tlef (read by the flow, 1_synth.log ODB-0227) defines DCCURRENTDENSITY / ACCURRENTDENSITY per via and metal layer, but no step compared the PSM currents (section 10) or signal/clock-net currents with them | the comparison, per layer and via, for power, signal and clock nets              | MISSING                                                                                    |
| Fault injection gaps                                                                                           | comparators, voter, clock, reset, inputs, MBU, common mode, fault_q 1 -> 0, thermal code 3, product in campaign, netlist/SDF injection, multi-seed                                                                                    | see 12.4; double-fault scenarios; netlist-level campaign                         | MISSING                                                                                    |
| Property checks the mutation run asks for                                                                      | m07, m20, m22, m27 survive                                                                                                                                                                                                            | assertions listed in section 14                                                  | MISSING                                                                                    |
| Formal on a netlist                                                                                            | all properties checked on RTL only                                                                                                                                                                                                    | rerun orbit_demo_fv.sv on a netlist                                              | MISSING                                                                                    |
| Regenerated published sep storage audit                                                                        | published reports quote 18/18 from a superseded script                                                                                                                                                                                | re-run make_pd-sep-report and republish                                          | MISSING (re-run only in this package)                                                      |
| Make target for sep multi-corner STA                                                                           | results exist only as a hand run                                                                                                                                                                                                      | pd-sep-corners target in mk/pdsep.mk publishing to reports/pdsep/                | MISSING                                                                                    |
| Counterexample / cover VCDs, per-trial campaign log                                                            | only in gitignored build/                                                                                                                                                                                                             | copy build/formal, build/fault traces and trials.tsv                             | MISSING (not in package)                                                                   |
| LEC netlists, make_pd-sep console log, 5_1_grt.json, antenna logs, LVS database and reference CDL, 6_final.odt | were only in gitignored build/ (D-99)                                                                                                                                                                                                 | —                                                                                | resolved in package: <a href="#">additional_evidence/</a> , <a href="#">6_final.odt.gz</a> |
| ECC integration and physical data                                                                              | ECC is standalone                                                                                                                                                                                                                     | integration spec, RTL, layout run                                                | N/A for this layout                                                                        |
| Pad ring, IO cells, ESD, package checks                                                                        | the layout is a core block: 0 padcells ( <a href="#">6_report.json</a> line 60)                                                                                                                                                       | —                                                                                | N/A (no pad ring, no package)                                                              |
| Analog / mixed-signal verification                                                                             | no analog blocks                                                                                                                                                                                                                      | —                                                                                | N/A (digital only)                                                                         |
| Silicon measurement                                                                                            | not fabricated                                                                                                                                                                                                                        | —                                                                                | N/A (no silicon)                                                                           |